

Lab04_FernandaVincenzo

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Q1:

```
library(readxl)
```

Warning: pacote 'readxl' foi compilado no R versão 4.5.3

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
v dplyr      1.1.4      v readr      2.1.5  
v forcats    1.0.0      v stringr    1.5.1  
v ggplot2    3.5.2      v tibble     3.2.1  
v lubridate  1.9.4      v tidyr      1.3.1  
v purrr      1.0.4
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

Q2:

```
fname <- file.path("datasets/wdiexcel.xlsx")  
file.exists(fname)
```

```
[1] TRUE
```

Q3:

```
countries <- read_excel(fname, sheet = "Country")
```

Q4:

```
countries <- countries |>
  filter(`Short Name` %in% c(
    "Brazil",
    "China",
    "Spain",
    "France",
    "Norway"
  ))
```

Q5:

```
WDI <- read_excel(fname, sheet = "Data")
```

Q6:

```
WDI <- WDI |>
  select(
    'Country Code',
    'Indicator Code',
    '1960':'2018'
  )
```

##Q7:

Não estão no formato tidy, pois todos os anos estão como colunas próprias.

```
WDI <- WDI |>
  pivot_longer(
    cols = `1960`:`2018`,
    names_to = "YEAR",
    values_to = "value"
  )
```

##Q8:

```
is.integer(WDI$YEAR)
```

[1] FALSE

```
WDI <- WDI |>
  mutate(YEAR = as.integer(YEAR))
```

```
is.integer(WDI$YEAR)
```

[1] TRUE

##Q9:

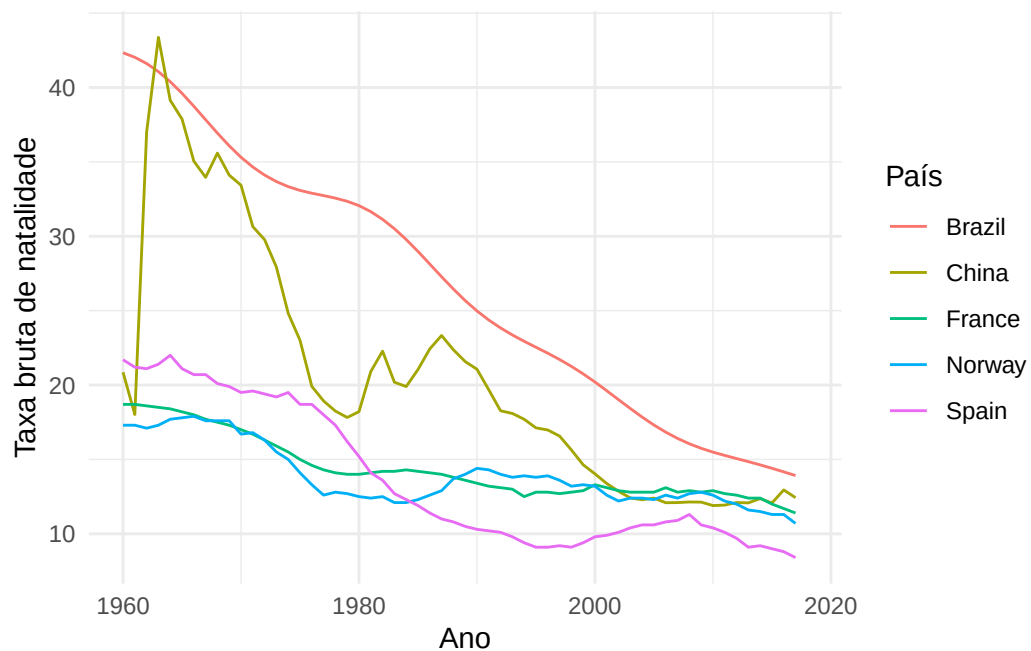
```
WDI <- WDI |>
  semi_join(
    countries,
    by = "Country Code"
  ) |>
  filter(`Indicator Code` == "SP.DYN.CBRT.IN") |>
  left_join(
    countries |>
      select(`Country Code`, `Short Name`),
    by = "Country Code"
  ) |>
  select(
    `Country Code`,
    `Short Name`,
    YEAR,
    `SP.DYN.CBRT.IN` = value
  )
```

##Q10:

```
WDI <- WDI |>
  arrange(`Short Name`, YEAR)
```

##Q11:

```
ggplot(
  WDI,
  aes(
    x = YEAR,
    y = `SP.DYN.CBRT.IN`,
    color = `Short Name`,
    group = `Short Name`
  )
) +
  geom_line(na.rm = TRUE) +
  labs(
    x = "Ano",
    y = "Taxa bruta de natalidade",
    color = "País"
  ) +
  theme_minimal()
```



```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-09-09 10:07:37 -03"
```